



# PROTOCOLO DE DIAGNÓSTICO DE ENFERMEDAD PERIODONTAL

Clínica Universitaria de Odontología

## 1. OBJETIVOS DEL DIAGNÓSTICO PERIODONTAL

El diagnóstico de las enfermedades periodontales busca:

- **Detectar** la presencia de la patología periodontal.
- **Clasificarla** cualitativa y cuantitativamente (estadiaje y grado).
- **Diseñar y proponer** un plan de tratamiento individualizado, basado en la evidencia científica actual.
- **Pronosticar** su evolución y establecer las pautas de mantenimiento a largo plazo.

## 2. ANAMNESIS Y EXPLORACIÓN GENERAL

- **Anamnesis completa:** Se deben recabar antecedentes médicos y odontológicos, incluyendo historia familiar de periodontitis.

Es fundamental identificar y documentar los factores de riesgo modificables y no modificables, como el tabaquismo (número de cigarrillos y años que lleva fumando), (es un momento ideal para indagar sobre la posibilidad de abandonar el hábito), la diabetes (control glucémico), el consumo de alcohol, el estrés crónico y el uso de ciertos medicamentos.

- **Motivo de consulta y enfermedad actual:** La inspección visual debe ser minuciosa, valorando el color, contorno y textura de las encías, la presencia de supuración, recesiones gingivales y fisuras de Stillman o hiperplasias. Se debe evaluar la calidad de la higiene oral del paciente.

### **3. EXPLORACIÓN PERIODONTAL COMPLETA**

- **Sondaje Periodontal:** Se utiliza una sonda periodontal sonda CP-12, con marcas a 3, 6, 9 y 12 mm) para registrar los siguientes parámetros en seis puntos por diente (mesio-bucal, bucal, disto-bucal, mesio-lingual/palatino, lingual/palatino, disto-lingual/palatino):
  - **Profundidad de Sondaje (PS):** Distancia desde el margen gingival hasta el fondo del surco.
  - **Recesión Gingival (RG):** Distancia desde la unión cemento-esmalte (LAC) hasta el margen gingival, medido en mm y reflejado en el periodontograma.
  - **Nivel de Inserción Clínica o Clinical Attachment Level (CAL):** Es el parámetro más importante para determinar la pérdida de inserción. Se calcula sumando la Profundidad de Sondaje (PS) y la Recesión Gingival
  - **Sangrado al Sondaje:** Se registra la presencia de sangrado en cada punto sondado. Se calcula el **Porcentaje de Sangrado al Sondaje**, que es el número de puntos que sangran dividido por el total de puntos sondados, multiplicado por 100.
  - **Supuración:** Se registra la presencia de supuración al sondear.
  - **Movilidad Dentaria:** Se clasifica del 0 al 3 (donde 0 es sin movilidad y 3 implica intrusión).

- **Índice de Placa de O'Leary:** Se registra la presencia o ausencia de placa en cuatro superficies por diente (distal, vestibular, mesial, lingual/palatino). Se calcula el porcentaje de superficies con placa.
- **Examen Radiográfico:** Es indispensable para evaluar la pérdida ósea. Se debe realizar una serie radiográfica periapical completa (18 radiografías) con la técnica del paralelismo, o una serie parcial, en función del número de dientes presentes. Esto permite valorar la pérdida de hueso horizontal y vertical y la presencia de afectación de furcas.
- **Registro gráfico** según la guía suministrada de "Registro de periodontogramas en Gesden". (ver protocolo de Gesden "Periodontograma" que se adjunta al final del protocolo)

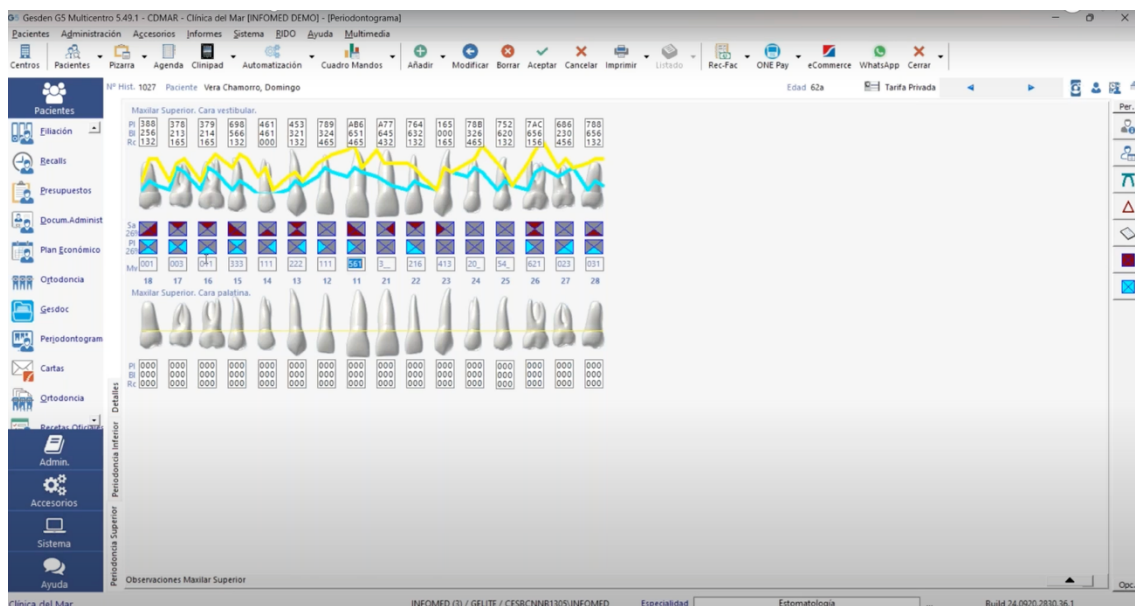


Figura 1: Periodontograma de Gesden

#### 4. DIAGNÓSTICO, CLASIFICACIÓN (ESTADIO Y GRADO)

La clasificación actual de la periodontitis (basada en la clasificación de 2017) se divide en **estadios** y **grados**.

##### **A. ESTADIAJE:**

Se basa en la **gravedad, extensión y complejidad** del caso.

- **Estadio I**
  - **CAL:** 1-2 mm.
  - **Pérdida ósea radiográfica:** <15% (limitada al tercio coronal de la raíz).
  - **Pérdida de dientes:** Ninguna.
- **Estadio II**
  - **(CAL):** 3-4 mm.
  - **Pérdida ósea radiográfica:** 15-33% (tercio coronal).
  - **Pérdida de dientes:** Ninguna pérdida de dientes por causa periodontal
- **Estadio III: Periodontitis Severa**
  - **CAL**  $\geq$  5 mm.
  - **Pérdida ósea radiográfica:**  $\geq$  33% (se extiende, o sobrepasa el tercio medio radicular).
  - **Pérdida de dientes:** Hasta 4 dientes perdidos por causa periodontal.
  - **Complejidad Alta.** Presencia de afectación de furca de grado II o III, defectos óseos verticales  $\geq$  3 mm, trauma oclusal.
- **Estadio IV: Periodontitis Avanzada**
  - **NIC (CAL):**  $\geq$  5 mm.

- **Pérdida ósea radiográfica:**  $\geq 33\%$  (se extiende al tercio medio o apical).
- **Pérdida de dientes:** 5 o más dientes perdidos por causa periodontal.
- **Complejidad muy alta:** Presencia de afectación de furca de grado II o III, defectos óseos verticales  $\geq 3$  mm. Además, hay disfunción masticatoria avanzada, como movilidad dentaria de grado 2 o 3, colapso de la mordida, o menos de 10 pares de dientes oponentes.

## ***B. GRADO:***

Se determina la velocidad de progresión de la enfermedad y el riesgo futuro.

- **Grado A: Progresión lenta**
  - **Criterio primario (preferible si hay posibilidad):** No hay evidencia de pérdida ósea en los últimos 5 años. Un criterio primario implica tener registros anteriores del paciente
  - **Criterio secundario:** Relación de la pérdida ósea (%) y la edad es  $<0.25$ .
  - **Factores de riesgo:** No fumador y ausencia de diabetes.
- **Grado B: Progresión moderada**
  - **Criterio primario:** Pérdida ósea  $< 2$  mm en los últimos 5 años.
  - **Criterio secundario:** Relación de la pérdida ósea (%) y la edad es  $0.25-1.0$ .
  - **Factores de riesgo:** Fumador de  $<10$  cigarrillos al día, diabetes con  $HbA1c <7.0\%$ .
- **Grado C: Progresión rápida**
  - **Criterio primario:** Pérdida ósea  $\geq 2$  mm en los últimos 5 años.
  - **Criterio secundario:** Relación de la pérdida ósea (%) y la edad es  $>1.0$ .

- **Factores de riesgo:** Fumador de  $\geq 10$  cigarrillos al día, diabetes con  $HbA1c \geq 7.0\%$ .

### **C. EXTENSIÓN:**

Se refiere a la distribución de la enfermedad.

- **Localizada:** Afecta a menos del **30%** de los dientes.
- **Generalizada:** Afecta al **30% o más** de los dientes.
- **Patrón molar/incisivo:** Tipo de periodontitis que afecta predominantemente a molares e incisivos.

### **NOTA: Recomendaciones y Aclaraciones**

- Es crucial que el estudiante entienda que la pérdida de inserción clínica (CAL) es el parámetro clave, ya que la profundidad de sondaje puede variar con la inflamación. El CAL es una medida más estable de la pérdida de inserción.
- Se debe enfatizar la importancia de la fuerza de sondaje (aproximadamente 0.2 N) para evitar mediciones inexactas
- La serie periapical completa es el "gold standard" para el diagnóstico. El estudiante debe aprender a interpretar los signos radiográficos de la periodontitis, identificando defectos óseos horizontales y verticales
- La evaluación del grado se prioriza mediante la observación longitudinal (comparando radiografías de diferentes momentos). Si no están disponibles, se utiliza el criterio secundario de la relación pérdida ósea/edad. Los factores de riesgo como el tabaquismo y la diabetes **modifican el grado**. De modo que, en principio, todos los pacientes son grado B, pero se deben buscar parámetros que eleven el grado a C (fumar más de 10 cigarrillos/día o  $HbA1c \geq 7\%$ ). Solo será grado

A un paciente sin progresión de la enfermedad que no fuma y no es diabético

## ESQUEMA PARA EL DIAGNOSTICO DE LA ENFERMADAD PERIDONTAL

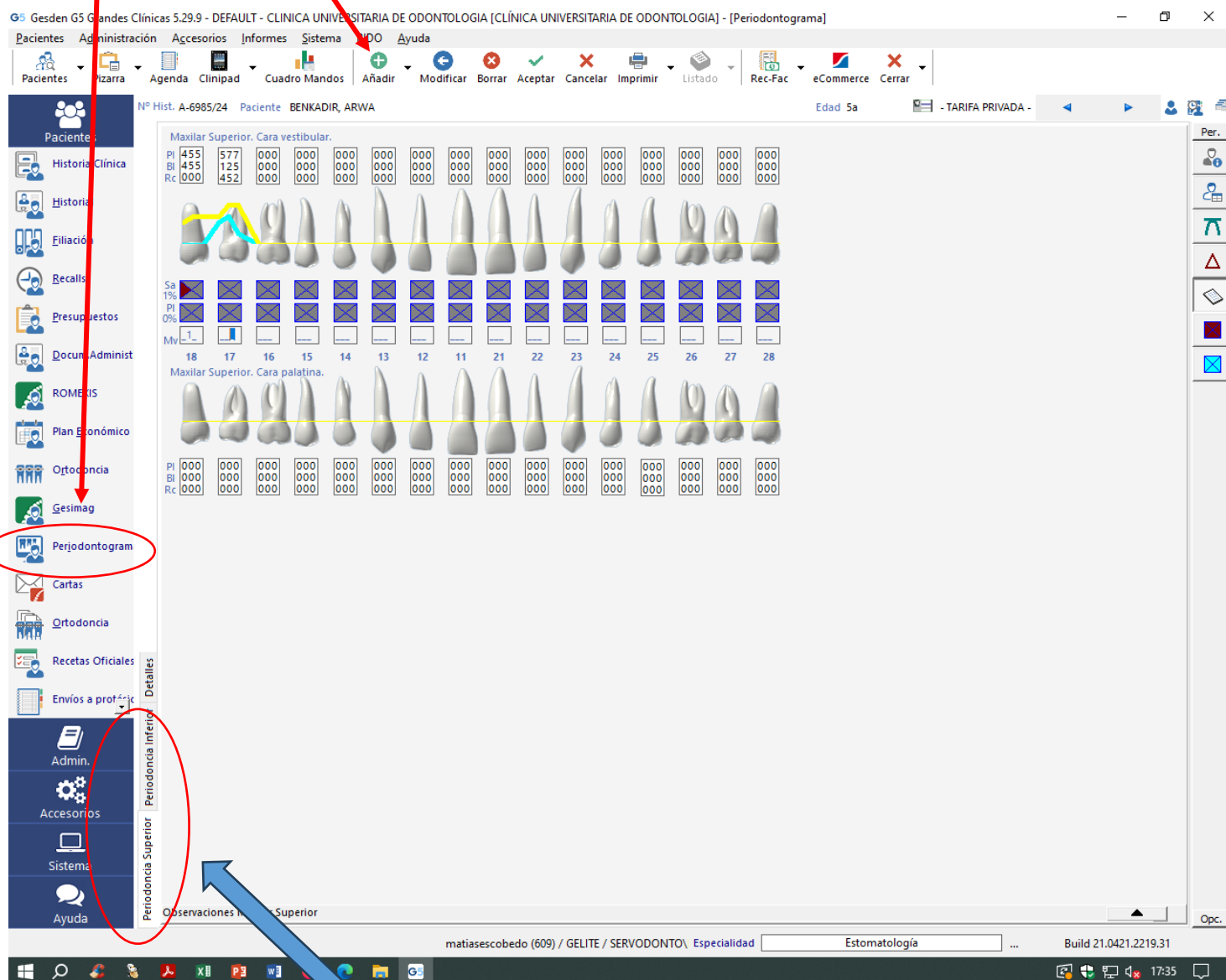
| Estadio de la Periodontitis |                                                 | Estadio I                                                                                                                                                                        | Estadio II                                                                                                                                                                       | Estadio III                                                                                                                                                                                                                                                               | Estadio IV                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------------------|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Severidad                   | PIC interproximal en el sitio con mayor pérdida | 1 a 2 mm                                                                                                                                                                         | 3 a 4 mm                                                                                                                                                                         | $\geq 5$ mm                                                                                                                                                                                                                                                               | $\geq 5$ mm                                                                                                                                                                                                                                                                                                                                                                                                |
|                             | Pérdida ósea radiográfica                       | Tercio coronal (<15%).                                                                                                                                                           | Tercio coronal (15-33%).                                                                                                                                                         | Se extiende al tercio medio o apical de la raíz.                                                                                                                                                                                                                          | Se extiende al tercio medio o apical de la raíz.                                                                                                                                                                                                                                                                                                                                                           |
|                             | Pérdida de dientes                              | Sin pérdida de dientes por periodontitis.                                                                                                                                        |                                                                                                                                                                                  | Pérdida de hasta 4 dientes debido a periodontitis.                                                                                                                                                                                                                        | Pérdida de 5 o más dientes debido a periodontitis.                                                                                                                                                                                                                                                                                                                                                         |
| Complejidad                 | Local                                           | <ul style="list-style-type: none"> <li>Máxima profundidad de sondaje <math>\leq 4</math>mm.</li> <li>En su mayoría la pérdida ósea se presenta con patrón horizontal.</li> </ul> | <ul style="list-style-type: none"> <li>Máxima profundidad de sondaje <math>\leq 5</math>mm.</li> <li>En su mayoría la pérdida ósea se presenta con patrón horizontal.</li> </ul> | En adición al estadio II: <ul style="list-style-type: none"> <li>Profundidad de sondaje <math>\geq 6</math>mm.</li> <li>Pérdida ósea vertical <math>\geq 3</math>mm.</li> <li>Comprometimiento de Furca clase II o III.</li> <li>Defecto moderado del reborde.</li> </ul> | En adición al estadio III:           Necesidad de rehabilitación compleja debido a: <ul style="list-style-type: none"> <li>Disfunción masticatoria.</li> <li>Trauma oclusal secundario (movilidad dentaria de grado <math>\geq 2</math>).</li> <li>Defecto severo del reborde.</li> <li>Problemas masticatorios, colapso oclusal Menos de 20 dientes remanescientes (10 pares de antagonistas).</li> </ul> |
| Distribución y extensión    | Añadir al estadio como descriptor               | Para cada estadio, se describe la extensión como localizada (<30% de los dientes involucrados) y generalizada (>30% de los dientes involucrados) o patrón molar/incisivo.        |                                                                                                                                                                                  |                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                            |

| Grado de la Periodontitis |                                   |                                                        | Grado A: progresión lenta                                     | Grado B: progresión moderada                                | Grado C: rápida progresión                                                                                                                                                                                                                                                                    |
|---------------------------|-----------------------------------|--------------------------------------------------------|---------------------------------------------------------------|-------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Criterios Primarios       | Evidencia directa de progresión   | Datos longitudinales (pérdida ósea radiográfica y PIC) | Evidencia de que no existe pérdida durante 5 años             | <2 mm en 5 años                                             | $\geq 2$ mm en 5 años                                                                                                                                                                                                                                                                         |
|                           | Evidencia indirecta de progresión | % de pérdida ósea/edad                                 | <0.25                                                         | 0.25 hasta 1.0                                              | >1.0                                                                                                                                                                                                                                                                                          |
|                           |                                   | Presentación Fenotípica                                | Gran acumulación de biofilm, con bajos niveles de destrucción | Destrucción periodontal compatible con depósitos de biofilm | La destrucción periodontal supera las expectativas por el acúmulo de biofilm. Parámetros clínicos específicos sugieren periodos de progresión rápida y/o afección temprana de la enfermedad (p. ej. Patrón molar/incisivo y falta de respuesta esperada a las terapias de control de biofilm) |
| Modificadores del Grado   | Factores de riesgo                | Tabaquismo                                             | No fumante                                                    | Fumante con <10 cigarrillos por día                         | Fumante con $\geq 10$ cigarrillos por día                                                                                                                                                                                                                                                     |
|                           |                                   | Diabetes                                               | Normoglicémico / sin evidencia diagnóstica de diabetes        | HbA1c <7.0% en pacientes diabéticos                         | HbA1c $\geq 7.0\%$ en pacientes diabéticos                                                                                                                                                                                                                                                    |



## GUÍA PARA REALIZAR UN PERIODONTOGRAMA EN GESDEN

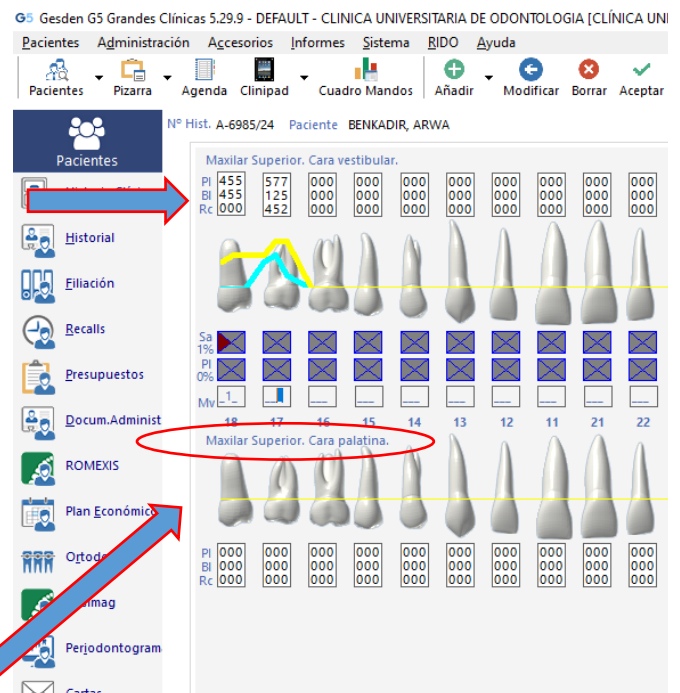
1º Debemos de clicar en periodontograma. Si ya tenemos uno realizado previamente y queremos hacer uno nuevo debemos de clicar en añadir.



2º El periodontograma empieza por defecto en el maxilar superior si solo quisiéramos hacer el inferior clicaríamos en el inferior.



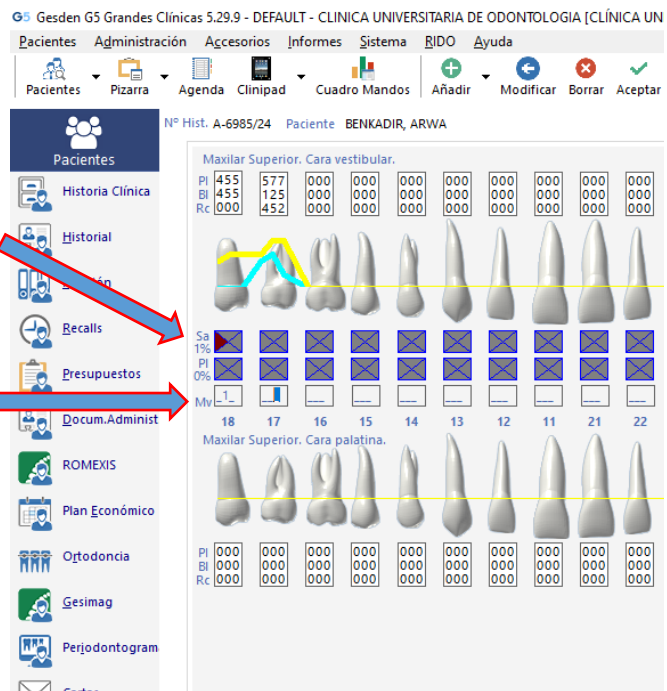
**3º Para empezar a marcar el Sondaje deberemos clicar en la fila intermedia de ceros (BI) y si quisiéramos indicar las posibles recesiones clicamos en la fila de ceros inferior (Rc). La fila superior de ceros es la Suma del BI + Rc.**



**Todo esto lo tenemos que repetir en la Cara palatina.**

**4º Para indicar el sangrado iremos marcando en cada diente las zonas de sangrado**

**5º Para la movilidad la Indicaremos en el apartado (Mv) con un número**



**Nota : El número máximo de sondaje que permite este programa es de 9mm si hubiera un sondaje superior habría que añadir “una nota” ( ver siguiente hoja) en el diente indicando el sondaje superior a 9mm**





# Periodontitis: Consensus report of workgroup 2 of the 2017 World Workshop on the Classification of Periodontal and Peri-Implant Diseases and Conditions

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The proceedings of the workshop were jointly and simultaneously published in the *Journal of Periodontology* and *Journal of Clinical Periodontology*.

**Abstract**

A new periodontitis classification scheme has been adopted, in which forms of the disease previously recognized as “chronic” or “aggressive” are now grouped under a single category (“periodontitis”) and are further characterized based on a multi-dimensional *staging* and *grading* system. *Staging* is largely dependent upon the severity of disease at presentation as well as on the complexity of disease management, while *grading* provides supplemental information about biological features of the disease including a history-based analysis of the rate of periodontitis progression; assessment of the risk for further progression; analysis of possible poor outcomes of treatment; and assessment of the risk that the disease or its treatment may negatively affect the general health of the patient.

Necrotizing periodontal diseases, whose characteristic clinical phenotype includes typical features (papilla necrosis, bleeding, and pain) and are associated with host immune response impairments, remain a distinct periodontitis category.

Endodontic-periodontal lesions, defined by a pathological communication between the pulpal and periodontal tissues at a given tooth, occur in either an acute or a chronic form, and are classified according to signs and symptoms that have direct impact on their prognosis and treatment.

Periodontal abscesses are defined as acute lesions characterized by localized accumulation of pus within the gingival wall of the periodontal pocket/sulcus, rapid tissue destruction and are associated with risk for systemic dissemination.

**KEYWORDS**

acute periodontal conditions, endo-periodontal lesion, necrotizing gingivitis, necrotizing periodontitis, periodontal abscess, periodontal disease, periodontitis

Periodontitis is a chronic multifactorial inflammatory disease associated with dysbiotic plaque biofilms and characterized by progressive destruction of the tooth-supporting apparatus. Its primary features include the loss of periodontal tissue support, manifested through clinical attachment loss (CAL) and radiographically assessed alveolar bone loss, presence of periodontal pocketing and gingival bleeding. Periodontitis is a major public health problem due to its high prevalence, as well as because it may lead to tooth loss and disability, negatively affect chewing function and aesthetics, be a source of social inequality, and impair quality of life. Periodontitis accounts for a substantial proportion of edentulism and masticatory dysfunction, results in significant dental care costs and has a plausible negative impact on general health.

According to the latest internationally accepted classification scheme (Armitage<sup>1</sup> 1999), *periodontitis* is further subdivided as follows:

- *Chronic periodontitis*, representing the forms of destructive periodontal disease that are generally characterized by slow progression
- *Aggressive periodontitis*, a diverse group of highly destructive forms of periodontitis affecting primarily young individuals,

including conditions formerly classified as “early-onset periodontitis” and “rapidly progressing periodontitis”

- *Periodontitis as a manifestation of systemic disease*, a heterogeneous group of systemic pathological conditions that include periodontitis as a manifestation
- *Necrotizing periodontal diseases*, a group of conditions that share a characteristic phenotype where necrosis of the gingival or periodontal tissues is a prominent feature
- *Periodontal abscesses*, a clinical entity with distinct diagnostic features and treatment requirements

Although the above classification has provided a workable framework that has been used extensively in both clinical practice and scientific investigation in periodontology during the past 17 years, the system suffers from several important shortcomings, including substantial overlap and lack of clear pathology-based distinction between the stipulated categories, diagnostic imprecision, and implementation difficulties. The objectives of workgroup 2 were to revisit the current classification system of periodontitis, incorporate new knowledge relevant to its epidemiology, etiology and pathogenesis that has accumulated since the current classification's inception, and propose a

new classification framework along with case definitions. To this end, five position papers were commissioned, authored, peer-reviewed, and accepted. The first reviewed the classification and diagnosis of aggressive periodontitis (Fine et al.<sup>2</sup> 2018); the second focused on the age-dependent distribution of clinical attachment loss in two population-representative, cross-sectional studies (Billings et al.<sup>3</sup> 2018); the third reviewed progression data of clinical attachment loss from existing prospective, longitudinal studies (Needleman et al.<sup>4</sup> 2018); the fourth reviewed the diagnosis, pathobiology, and clinical presentation of acute periodontal lesions (periodontal abscesses, necrotizing periodontal diseases and endo-periodontal lesions; Herrera et al.<sup>5</sup> 2018); lastly, the fifth focused on periodontitis case definitions (Tonetti et al.<sup>6</sup> 2018), Table 1.

The workgroup reviewed, debated and agreed by consensus on the overall conclusions of the five position papers, that can be largely summarized as follows:

1. The conflicting literature findings on aggressive periodontitis are primarily due to the fact that (i) the currently adopted classification is too broad, (ii) the disease has not been studied from its inception, and (iii) there is paucity of longitudinal studies including multiple time points and different populations. The position paper argued that a more restrictive definition might be better suited to take advantage of modern methodologies to enhance knowledge on the diagnosis, pathogenesis, and management of this form of periodontitis.

**TABLE 1A** Classification of periodontitis based on stages defined by severity (according to the level of interdental clinical attachment loss, radiographic bone loss and tooth loss), complexity and extent and distribution

| Periodontitis stage     |                                          | Stage I                                                                                                      | Stage II                                                   | Stage III                                                                                                                                                | Stage IV                                                                                                                                                                                                                                                                          |
|-------------------------|------------------------------------------|--------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Severity                | Interdental CAL at site of greatest loss | 1 to 2 mm                                                                                                    | 3 to 4 mm                                                  | ≥5 mm                                                                                                                                                    | ≥5 mm                                                                                                                                                                                                                                                                             |
|                         | Radiographic bone loss                   | Coronal third (<15%)                                                                                         | Coronal third (15% to 33%)                                 | Extending to middle or apical third of the root                                                                                                          | Extending to middle or apical third of the root                                                                                                                                                                                                                                   |
|                         | Tooth loss                               | No tooth loss due to periodontitis                                                                           |                                                            | Tooth loss due to periodontitis of ≤4 teeth                                                                                                              | Tooth loss due to periodontitis of ≥5 teeth                                                                                                                                                                                                                                       |
| Complexity              | Local                                    | Maximum probing depth ≤4 mm<br>Mostly horizontal bone loss                                                   | Maximum probing depth ≤5 mm<br>Mostly horizontal bone loss | In addition to stage II complexity:<br>Probing depth ≥6 mm<br>Vertical bone loss ≥3 mm<br>Furcation involvement Class II or III<br>Moderate ridge defect | In addition to stage III complexity:<br>Need for complex rehabilitation due to:<br>Masticatory dysfunction<br>Secondary occlusal trauma (tooth mobility degree ≥2)<br>Severe ridge defect<br>Bite collapse, drifting, flaring<br>Less than 20 remaining teeth (10 opposing pairs) |
|                         |                                          |                                                                                                              |                                                            |                                                                                                                                                          |                                                                                                                                                                                                                                                                                   |
| Extent and distribution | Add to stage as descriptor               | For each stage, describe extent as localized (<30% of teeth involved), generalized, or molar/incisor pattern |                                                            |                                                                                                                                                          |                                                                                                                                                                                                                                                                                   |

The initial stage should be determined using clinical attachment loss (CAL); if not available then radiographic bone loss (RBL) should be used. Information on tooth loss that can be attributed primarily to periodontitis – if available – may modify stage definition. This is the case even in the absence of complexity factors. Complexity factors may shift the stage to a higher level, for example furcation II or III would shift to either stage III or IV irrespective of CAL. The distinction between stage III and stage IV is primarily based on complexity factors. For example, a high level of tooth mobility and/or posterior bite collapse would indicate a stage IV diagnosis. For any given case only some, not all, complexity factors may be present, however, in general it only takes one complexity factor to shift the diagnosis to a higher stage. It should be emphasized that these case definitions are guidelines that should be applied using sound clinical judgment to arrive at the most appropriate clinical diagnosis.

For post-treatment patients, CAL and RBL are still the primary stage determinants. If a stage-shifting complexity factor(s) is eliminated by treatment, the stage should not retrogress to a lower stage since the original stage complexity factor should always be considered in maintenance phase management.



**TABLE 1B** Classification of periodontitis based on grades that reflect biologic features of the disease including evidence of, or risk for, rapid progression, anticipated treatment response, and effects on systemic health

| Periodontitis grade |                                     |                                                            | Grade A:<br>Slow rate of<br>progression                        | Grade B:<br>Moderate rate of<br>progression             | Grade C:<br>Rapid rate of<br>progression                                                                                                                                                                                                                                                |
|---------------------|-------------------------------------|------------------------------------------------------------|----------------------------------------------------------------|---------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Primary criteria    | Direct evidence of<br>progression   | Longitudinal data<br>(radiographic<br>bone loss or<br>CAL) | Evidence of no loss<br>over 5 years                            | <2 mm over 5 years                                      | ≥2 mm over 5 years                                                                                                                                                                                                                                                                      |
|                     | Indirect evidence of<br>progression | % bone loss/age                                            | <0.25                                                          | 0.25 to 1.0                                             | >1.0                                                                                                                                                                                                                                                                                    |
|                     |                                     | Case phenotype                                             | Heavy biofilm<br>deposits with low<br>levels of<br>destruction | Destruction<br>commensurate<br>with biofilm<br>deposits | Destruction exceeds<br>expectation given biofilm<br>deposits; specific clinical<br>patterns suggestive of<br>periods of rapid<br>progression and/or early<br>onset disease (e.g.,<br>molar/incisor pattern;<br>lack of expected response<br>to standard bacterial<br>control therapies) |
| Grade modifiers     | Risk factors                        | Smoking                                                    | Non-smoker                                                     | Smoker <10<br>cigarettes/day                            | Smoker ≥10 cigarettes/day                                                                                                                                                                                                                                                               |
|                     |                                     | Diabetes                                                   | Normoglycemic / no<br>diagnosis of<br>diabetes                 | HbA1c <7.0% in<br>patients with<br>diabetes             | HbA1c ≥7.0% in patients<br>with diabetes                                                                                                                                                                                                                                                |

Grade should be used as an indicator of the rate of periodontitis progression. The primary criteria are either direct or indirect evidence of progression. Whenever available, direct evidence is used; in its absence indirect estimation is made using bone loss as a function of age at the most affected tooth or case presentation (radiographic bone loss expressed as percentage of root length divided by the age of the subject, RBL/age). Clinicians should initially assume grade B disease and seek specific evidence to shift towards grade A or C, if available. Once grade is established based on evidence of progression, it can be modified based on the presence of risk factors. CAL = clinical attachment loss; HbA1c = glycated hemoglobin A1c; RBL = radiographic bone loss.

- Despite substantial differences in the overall severity of attachment loss between the two population samples analyzed by Billings et al.<sup>3</sup>, suggesting presence of cohort effects, common patterns of CAL were identified across different ages, along with consistencies in the relative contribution of recession and pocket depth to CAL. The findings suggest that it is feasible to introduce empirical evidence-driven thresholds of attachment loss that signify disproportionate severity of periodontitis with respect to age.
- Longitudinal mean annual attachment level change was found to vary considerably both within and between populations. Surprisingly, neither age nor sex had any discernible effects on CAL change, but geographic location was associated with differences. Overall, the position paper argued that the existing evidence neither supports nor refutes the differentiation between forms of periodontal diseases based upon progression of attachment level change.
- Necrotizing periodontal diseases are characterized by three typical clinical features (papilla necrosis, bleeding, and pain) and are associated with host immune response impairments,

which should be considered in the classification of these conditions (Table 2).

Endodontic-periodontal lesions are defined by a pathological communication between the pulpal and periodontal tissues at a given tooth, occur in either an acute or a chronic form, and should be classified according to signs and symptoms that have direct impact on their prognosis and treatment (i.e., presence or absence of fractures and perforations, and presence or absence of periodontitis) (Table 3).

Periodontal abscesses most frequently occur in pre-existing periodontal pockets and should be classified according to their etiology. They are characterized by localized accumulation of pus within the gingival wall of the periodontal pocket/sulcus, cause rapid tissue destruction which may compromise tooth prognosis, and are associated with risk for systemic dissemination (Table 4).

- A periodontitis case definition system should include three components: (a) identification of a patient as a periodontitis case, (b) identification of the specific type of periodontitis, and (c) description of

**TABLE 2** Classification of necrotizing periodontal diseases (NPD)

| Category                                                                               | Patients                  | Predisposing conditions                                   | Clinical condition                         |
|----------------------------------------------------------------------------------------|---------------------------|-----------------------------------------------------------|--------------------------------------------|
| Necrotizing periodontal diseases in chronically, severely compromised patients         | In adults                 | HIV+/AIDS with CD4 counts < 200 and detectable viral load | NG, NP, NS, Noma. Possible progression     |
|                                                                                        |                           | Other severe systemic conditions (immunosuppression)      |                                            |
|                                                                                        | In children               | Severe malnourishments <sup>a</sup>                       |                                            |
|                                                                                        |                           | Extreme living conditions <sup>b</sup>                    |                                            |
|                                                                                        |                           | Severe (viral) infections <sup>c</sup>                    |                                            |
| Necrotizing periodontal diseases in temporarily and/or moderately compromised patients | In gingivitis patients    | Uncontrolled factors: stress, nutrition, smoking, habits  | Generalized NG. Possible progression to NP |
|                                                                                        |                           | Previous NPD: residual craters                            | Localized NG. Possible progression to NP   |
|                                                                                        |                           | Local factors: root proximity, tooth malposition          |                                            |
|                                                                                        | In periodontitis patients | Common predisposing factors for NPD (see paper)           | NG. Infrequent progression                 |
|                                                                                        |                           |                                                           | NP. Infrequent progression                 |

NG, necrotizing gingivitis; NP, necrotizing periodontitis; NS, necrotizing stomatitis.

<sup>a</sup>Mean plasma and serum concentrations of retinol, total ascorbic acid, zinc, and albumin markedly reduced, or very marked depletion of plasma retinol, zinc, and ascorbate; and saliva levels of albumin and cortisol, as well as plasma cortisol concentrations, significantly increased.

<sup>b</sup>Living in substandard accommodations, exposure to debilitating childhood diseases, living near livestock, poor oral hygiene, limited access to potable water and poor sanitary disposal of human and animal fecal waste.

<sup>c</sup>Measles, herpes viruses (cytomegalovirus, Epstein-Barr virus-1, herpes simplex virus), chicken pox, malaria, febrile illness.

**TABLE 3** Classification of endo-periodontal lesions

|                                             |                                                       |                                                             |
|---------------------------------------------|-------------------------------------------------------|-------------------------------------------------------------|
| Endo-periodontal lesion with root damage    | Root fracture or cracking                             |                                                             |
|                                             | Root canal or pulp chamber perforation                |                                                             |
|                                             | External root resorption                              |                                                             |
| Endo-periodontal lesion without root damage | Endo-periodontal lesion in periodontitis patients     | Grade 1 – narrow deep periodontal pocket in 1 tooth surface |
|                                             |                                                       | Grade 2 – wide deep periodontal pocket in 1 tooth surface   |
|                                             |                                                       | Grade 3 – deep periodontal pockets in > 1 tooth surface     |
|                                             | Endo-periodontal lesion in non-periodontitis patients | Grade 1 – narrow deep periodontal pocket in 1 tooth surface |
|                                             |                                                       | Grade 2 – wide deep periodontal pocket in 1 tooth surface   |
|                                             |                                                       | Grade 3 – deep periodontal pockets in > 1 tooth surface     |

the clinical presentation and other elements that affect clinical management, prognosis, and potentially broader influences on both oral and systemic health. A framework for developing a multi-dimensional periodontitis *staging* and *grading* system was proposed, in which *staging* (Table 1A) is largely dependent upon the severity of disease at presentation as well as on the complexity of disease management, while *grading* (Table 1B) provides supplemental information about biological features of the disease including a history-based

analysis of the rate of periodontitis progression; assessment of the risk for further progression; analysis of possible poor outcomes of treatment; and assessment of the risk that the disease or its treatment may negatively affect the general health of the patient.

During the workgroup deliberations, the following questions were formulated and addressed in order to clarify and substantiate the need for a new classification system for periodontitis:



**TABLE 4** Classification of periodontal abscesses based on the etiologic factors involved

|                                                                                                             |                            |                                         |                                                                            |
|-------------------------------------------------------------------------------------------------------------|----------------------------|-----------------------------------------|----------------------------------------------------------------------------|
| Periodontal abscess in periodontitis patients (in a pre-existing periodontal pocket)                        | Acute exacerbation         | Untreated periodontitis                 |                                                                            |
|                                                                                                             |                            | Non-responsive to therapy periodontitis |                                                                            |
|                                                                                                             |                            | Supportive periodontal therapy          |                                                                            |
|                                                                                                             | After treatment            | Post-scaling                            | Systemic antimicrobials<br>Other drugs: nifedipine                         |
|                                                                                                             |                            | Post-surgery                            |                                                                            |
|                                                                                                             |                            | Post-medication                         |                                                                            |
| Periodontal abscess in non-periodontitis patients (not mandatory to have a pre-existing periodontal pocket) | Impaction                  |                                         | Dental floss, orthodontic elastic, toothpick, rubber dam, or popcorn hulls |
|                                                                                                             | Harmful habits             |                                         | Wire or nail biting and clenching                                          |
|                                                                                                             | Orthodontic factors        |                                         | Orthodontic forces or a cross-bite                                         |
|                                                                                                             | Gingival overgrowth        |                                         |                                                                            |
|                                                                                                             | Alteration of root surface | Severe anatomic alterations             | Invaginated tooth, dens evaginatus or odontodysplasia                      |
|                                                                                                             |                            | Minor anatomic alterations              | Cemental tears, enamel pearls or developmental grooves                     |
|                                                                                                             |                            | Iatrogenic conditions                   | Perforations                                                               |
|                                                                                                             |                            | Severe root damage                      | Fissure or fracture, cracked tooth syndrome                                |
|                                                                                                             |                            | External root resorption                |                                                                            |

### Which are the main features that identify periodontitis?

Loss of periodontal tissue support due to inflammation is the primary feature of periodontitis. A threshold of interproximal, CAL of  $\geq 2$  mm or  $\geq 3$  mm at  $\geq 2$  non-adjacent teeth is commonly used. Clinicians typically confirm presence of interproximal tissue loss through radiographic assessments of bone loss. Clinically meaningful descriptions of periodontitis should include the proportion of sites that bleed on probing, and the number and proportion of teeth with probing depth over certain thresholds (commonly  $\geq 4$  mm and  $\geq 6$  mm) and of teeth with CAL of  $\geq 3$  mm and  $\geq 5$  mm (Holtfreter et al.<sup>7</sup>).

### Which criteria would need to be fulfilled to support the contention that chronic and aggressive periodontitis are indeed different diseases? (e.g., etiology, histology, pathophysiology, clinical presentation, other)

Differences in etiology and pathophysiology are required to indicate presence of distinct periodontitis entities; variations in clinical presentation *per se*, i.e. extent and severity, do not support the concept of different diseases.

### Does current evidence suggest that we should continue to differentiate between “aggressive” and “chronic” periodontitis as two different diseases?

Current evidence does not support the distinction between chronic and aggressive periodontitis, as defined by the 1999 Classification Workshop, as two separate diseases; however, a substantial variation in clinical presentation exists with respect to extent and severity throughout the age spectrum, suggesting that there are population subsets with distinct disease trajectories due to differences in exposure and/or susceptibility.

### Is there evidence suggesting that early-onset forms of periodontitis (currently classified under “aggressive periodontitis”) have a distinct pathophysiology (e.g., genetic background, microbiology, host-response) compared to later-onset forms?

Although localized early onset periodontitis has a distinct, well-recognized clinical presentation (early onset, molar/incisor distribution, progression of attachment loss), the specific etiologic or pathological elements that account for this distinct presentation are insufficiently

defined. Likewise, mechanisms accounting for the development of generalized periodontitis in young individuals are poorly understood.

### What are the determinants for the mean annual attachment loss based on existing longitudinal studies in adults?

A meta-analysis included in the position paper documented differences in mean annual attachment loss between studies originating from different geographic regions but did not reveal an association with age or sex. It should be emphasized that meta-analysis of mean data may fail to identify associations due to the loss of information and the lack of accounting for both disease progression and regression. However, approaches that have modelled both progression and regression of CAL have also reported no effect of age or smoking on progression, although age and smoking reduced disease regression (e.g., Faddy et al.<sup>8</sup>). Individual studies that could not be included in the meta-analysis have shown effects of smoking, socioeconomic status, previous attachment loss, ethnicity, age, sex, and calculus on mean annual attachment loss.

### How do we define a patient as a periodontitis case?

In the context of clinical care, a patient is a "periodontitis case" if:

1. Interdental CAL is detectable at  $\geq 2$  non-adjacent teeth, or
2. Buccal or oral CAL  $\geq 3$  mm with pocketing  $\geq 3$  mm is detectable at  $\geq 2$  teeth but the observed CAL cannot be ascribed to non-periodontitis-related causes such as: 1) gingival recession of traumatic origin; 2) dental caries extending in the cervical area of the tooth; 3) the presence of CAL on the distal aspect of a second molar and associated with malposition or extraction of a third molar, 4) an endodontic lesion draining through the marginal periodontium; and 5) the occurrence of a vertical root fracture.

### Which different forms of periodontitis are recognized in the present revised classification system?

Based on pathophysiology, three clearly different forms of periodontitis have been identified:

- (A) Necrotizing periodontitis
- (B) Periodontitis as a direct manifestation of systemic diseases
- (C) Periodontitis

Differential diagnosis is based on history and the specific signs and symptoms of necrotizing periodontitis, or the presence or absence of an uncommon systemic disease that alters the host immune response. Periodontitis as a direct manifestation of systemic disease (Albandar et al.<sup>9</sup>, Jepsen et al.<sup>10</sup>) should follow the classification of the primary disease according to the respective International Statistical Classification of Diseases and Related Health Problems (ICD) codes.

The remaining clinical cases of periodontitis which do not have the local characteristics of necrotizing periodontitis or the systemic characteristics of a rare immune disorder with a secondary manifestation of periodontitis should be diagnosed as "periodontitis" and be further characterized using a staging and grading system that describes clinical presentation as well as other elements that affect clinical management, prognosis, and potentially broader influences on both oral and systemic health.

### How is a periodontitis case further characterized by stage and grade?

An individual case of periodontitis should be further characterized using a simple matrix that describes the *stage* and *grade* of the disease. Stage is largely dependent upon the severity of disease at presentation, as well as on the anticipated complexity of disease management, and further includes a description of extent and distribution of the disease in the dentition. Grade provides supplemental information about biological features of the disease including a history-based analysis of the rate of periodontitis progression; assessment of the risk for further progression; analysis of possible poor outcomes of treatment; and assessment of the risk that the disease or its treatment may negatively affect the general health of the patient. For a complete description of the rationale, determinants, and practical implementation of the staging and grading system, refer to Tonetti et al.<sup>6</sup> Tables 1 and 2 list the framework of the staging and grading system.

### Do the acute periodontal lesions have distinct features when compared with other forms of periodontitis?

Periodontal abscesses, lesions from necrotizing periodontal diseases and acute presentations of endo-periodontal lesions, share the following features that differentiate them from periodontitis lesions: (1) rapid-onset, (2) rapid destruction of periodontal tissues, underscoring the importance of prompt treatment, and (3) pain or discomfort, prompting patients to seek urgent care.

### Do periodontal abscesses have a distinct pathophysiology when compared to other periodontitis lesions?

The first step in the development of a periodontal abscess is bacterial invasion or foreign body impaction in the soft tissues surrounding the periodontal pocket, which develops into an inflammatory process that attracts polymorphonuclear neutrophils (PMNs) and low numbers of other immune cells. If the neutrophil-mediated defense process fails to control the local bacterial invasion or clear the foreign body, degranulation, necrosis and further neutrophilic influx may occur, leading to the formation of pus which, if not drained, results in an abscess. Pathophysiologically, this lesion differs in that the low pH within an abscess leads to rapid enzymatic disruption

of the surrounding connective tissues and, in contrast to a chronic inflammatory lesion, has a greater potential for resolution if quickly managed.

### What is the case definition of a periodontal abscess?

Periodontal abscess is a localized accumulation of pus located within the gingival wall of the periodontal pocket/sulcus, resulting in a significant tissue breakdown. The primary detectable signs/symptoms associated with a periodontal abscess may involve ovoid elevation in the gingiva along the lateral part of the root and bleeding on probing. Other signs/symptoms that may also be observed include pain, suppuration on probing, deep periodontal pocket, and increased tooth mobility.

A periodontal abscess may develop in a pre-existing periodontal pocket, e.g., in patients with untreated periodontitis, under supportive therapy or after scaling and root planing or systemic antimicrobial therapy. A periodontal abscess occurring at a previously periodontally healthy site is commonly associated with a history of impaction or harmful habits.

### Do necrotizing periodontal diseases have a distinct pathophysiology when compared to other periodontitis lesions?

Yes. Necrotizing gingivitis lesions are characterized by the presence of ulcers within the stratified squamous epithelium and the superficial layer of the gingival connective tissue, surrounded by a non-specific acute inflammatory infiltrate. Four zones have been described: (1) superficial bacterial zone, (2) neutrophil-rich zone, (3) necrotic zone and (4) a spirochetal/bacterial infiltration zone.

Necrotizing periodontal diseases are strongly associated with impairment of the host immune system, as follows: (1) in chronically, severely compromised patients (e.g., AIDS patients, children suffering from severe malnourishment, extreme living conditions, or severe infections) and may constitute a severe or even life-threatening condition; and (2) in temporarily and/or moderately compromised patients (e.g., in smokers or psycho-socially stressed adult patients).

### What are the case definitions of necrotizing periodontal diseases?

*Necrotizing gingivitis* is an acute inflammatory process of the gingival tissues characterized by presence of necrosis/ulcer of the interdental papillae, gingival bleeding, and pain. Other signs/symptoms associated with this condition may include halitosis, pseudomembranes, regional lymphadenopathy, fever, and sialorrhea (in children).

*Necrotizing periodontitis* is an inflammatory process of the periodontium characterized by presence of necrosis/ulcer of the interdental papillae, gingival bleeding, halitosis, pain, and rapid bone loss. Other signs/symptoms associated with this condition may include pseudomembrane formation, lymphadenopathy, and fever.

*Necrotizing stomatitis* is a severe inflammatory condition of the periodontium and the oral cavity in which soft tissue necrosis

extends beyond the gingiva and bone denudation may occur through the alveolar mucosa, with larger areas of osteitis and formation of bone sequestrum. It typically occurs in severely systemically compromised patients. Atypical cases have also been reported, in which necrotizing stomatitis may develop without prior appearance of necrotizing gingivitis/periodontitis lesions.

### Do endo-periodontal lesions have a distinct pathophysiology when compared to other periodontitis or endodontic lesions?

The term endo-periodontal lesion describes a pathologic communication between the pulpal and periodontal tissues at a given tooth that may be triggered by a carious or traumatic lesion that affects the pulp and, secondarily, affects the periodontium; by periodontal destruction that secondarily affects the root canal; or by concomitant presence of both pathologies. The review did not identify evidence for a distinct pathophysiology between an endo-periodontal and a periodontal lesion. Nonetheless, the communication between the pulp/root canal system and the periodontium complicates the management of the involved tooth.

### What is the case definition of an endo-periodontal lesion?

Endo-periodontal lesion is a pathologic communication between the pulpal and periodontal tissues at a given tooth that may occur in an acute or a chronic form. The primary signs associated with this lesion are deep periodontal pockets extending to the root apex and/or negative/altered response to pulp vitality tests. Other signs/symptoms may include radiographic evidence of bone loss in the apical or furcation region, spontaneous pain or pain on palpation/percussion, purulent exudate/suppuration, tooth mobility, sinus tract/fistula, and crown and/or gingival color alterations. Signs observed in endo-periodontal lesions associated with traumatic and/or iatrogenic factors may include root perforation, fracture/cracking, or external root resorption. These conditions drastically impair the prognosis of the involved tooth.

### Which are the current key gaps in knowledge that would inform a better classification of periodontitis and should be addressed in future research?

Future research should:

1. Develop improved methodologies to assess more accurately the longitudinal soft and hard tissue changes associated with periodontitis progression
2. Identify genetic, microbial, and host response-associated markers that differentiate between distinct periodontitis phenotypes, or which can reflect the initiation and progression of periodontitis.
3. Expand existing epidemiological databases to include world regions currently underrepresented, utilizing consistent, standardized methodologies, and capturing and reporting detailed data on

both patient-related, oral, and periodontal variables. Open access to the detailed data is crucial to facilitate comprehensive analyses.

4. Integrate multi-dimensional data platforms (clinical, radiographic, -omics) to facilitate systems biology approaches to the study of periodontal and peri-implant diseases and conditions
5. Use existing databases/ develop new databases that will facilitate the implementation, validation and continuous refinement of the newly introduced periodontitis classification system.

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**FIGURE 1** Participants of Workgroup 2